

Cast Film Production with Polyethylene Recycled from a Post-Industrial Printed Multilayer Film by Solvent-Targeted Recovery and Precipitation

Kevin L. Sánchez-Rivera, Charles Granger, Harrison Appiah, Kevin Nelson, Steve Grey, David J. Sun, John E. Estela-García, Edward Chen, Zhuo Xu, Tim A. Osswald, Lih-Sheng Turng, Armando G. McDonald, Reid C. Van Lehn, Ezra Bar-Ziv, and George W. Huber*



Cite This: *ACS Materials Lett.* 2024, 6, 4042–4050



Read Online

ACCESS |



Metrics & More

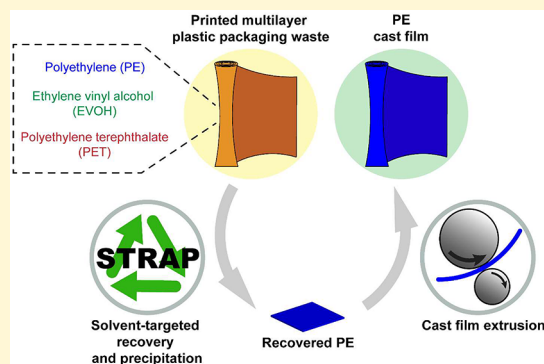


Article Recommendations



Supporting Information

ABSTRACT: The solvent-targeted recovery and precipitation (STRAP) process separates and recovers the constituent resins in multilayer plastic packaging films by selective polymer dissolution. In this work, three different experimental methods were considered to recover a polyethylene (PE) resin from a printed multilayer film by STRAP. The methods consisted of (1) a filter bag system, (2) a Soxhlet extraction, and (3) a jacketed dissolution vessel. Cast films were produced with the PE recovered from each method and were analyzed for color, mechanical properties, and number of impurities. High-quality recycled PE cast films can be produced by increasing the solvent to plastic ratio, including a filter pore size of 100 μm , and optimizing temperature control in STRAP. This study demonstrates that STRAP polymers can be recycled back into plastic films, enabling the potential circularity of these packaging materials.



The packaging industry is one of the largest markets for single-use plastics, generating more global plastic waste than the transportation, textiles, healthcare, and construction industries.^{1,2} Currently, plastic packaging is mostly landfilled or incinerated in the United States and efforts are needed to improve the circularity of these materials by addressing recycling challenges.³ Despite its end-of-life issues, flexible plastic packaging offers many benefits to consumers in food preservation. Flexible packaging has lower environmental impacts than other materials like glass, metals, or rigid plastic containers.^{4–6} These flexible plastic materials can be manufactured as monolayer, coated monolayer, or multilayer films, with about 17% of all films produced being multilayer films containing multiple polymer components.⁷ Typical polymers found in these structures include polyolefins like polyethylene (PE) or polypropylene (PP) for their moisture barrier properties, ethylene vinyl alcohol (EVOH) for its gas barrier properties, polyethylene terephthalate (PET) for its heat resistance and mechanical strength, modified ethylene vinyl acetate (EVA) or other tie layers for their ability to adhere dissimilar materials, among others.^{8,9} In most cases, these

multilayer materials are intended for single use and their recycling is challenging due to difficulties in the separation of their complex structures.^{10,11}

Solvent-Targeted Recovery and Precipitation (STRAP) recycles multilayer films by using a series of solvents to selectively recover the constituent resins of the material, guided by thermodynamic calculations of polymer solubility.^{12–15} The STRAP process can enable a recycling pathway for rigid and flexible multilayer films by recovering polymers like polyethylene (PE), ethylene vinyl alcohol (EVOH), and polyethylene terephthalate (PET) with >95 wt % material efficiency. This approach can produce polymers with comparable properties to pure resins, can be economically feasible, and can introduce environmental benefits when compared to film

Received: May 20, 2024

Revised: July 9, 2024

Accepted: July 9, 2024

Published: August 1, 2024



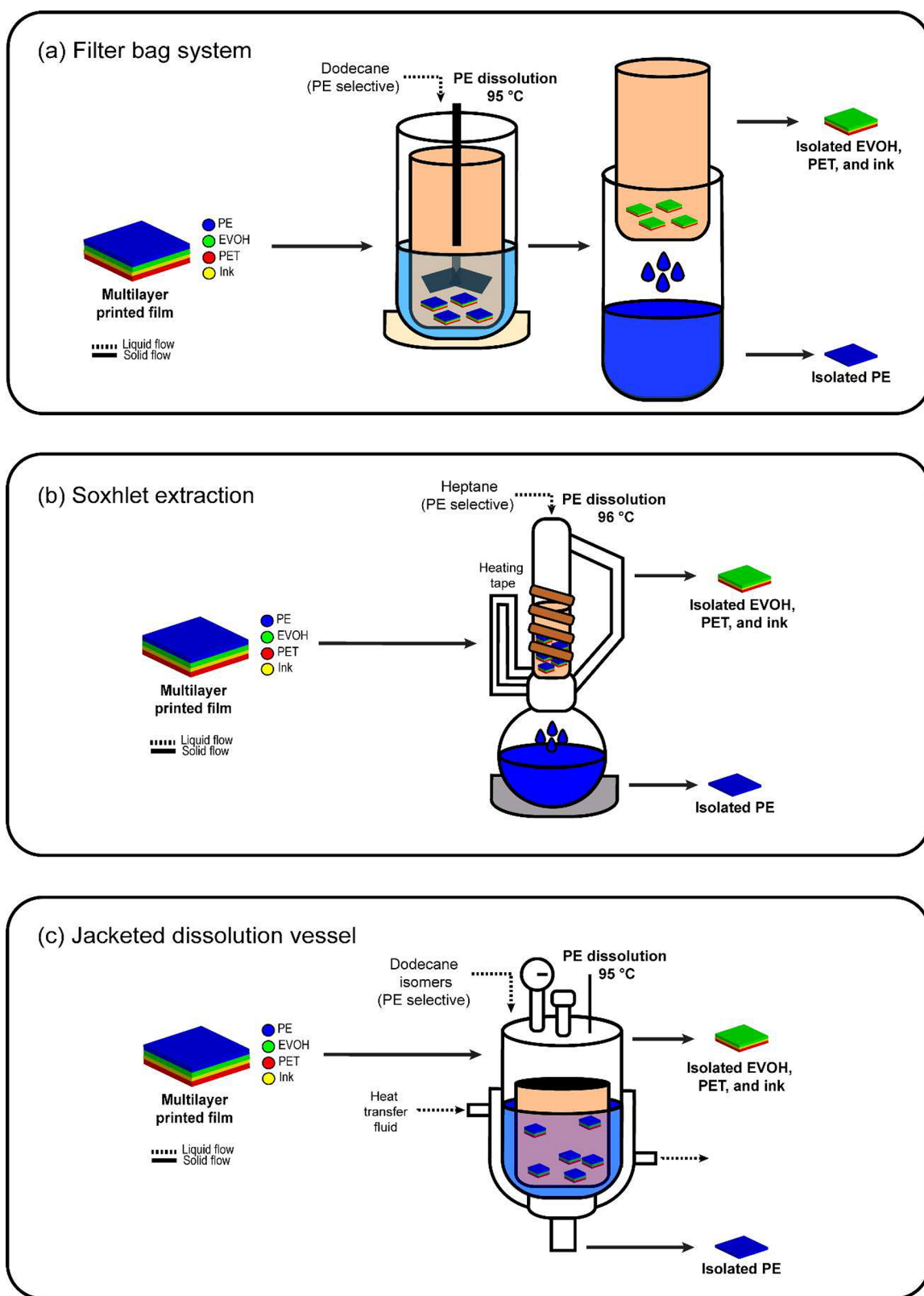


Figure 1. Recovery of polyethylene (PE) from a postindustrial printed multilayer film via three different solvent-targeted recovery and precipitation (STRAP) methods: (a) filter bag system, (b) Soxhlet extraction, and (c) jacketed dissolution vessel.

production from pure polymers.^{16–18} Dissolution-based recycling processes like STRAP have gained interest in recent years due to their potential for recovering valuable polymeric components from different types of plastic waste while removing

plastic additives, adhesives, impurities, odor, and color.^{19–24}

Companies like Polystyvert, PureCycle, and APK are commercializing dissolution-based processes.²⁵

Table 1. Experimental Parameters and PE Yields from the Three STRAP Methods

	Method	Solvent	Feed (g)	Solvent volume (L)	Solvent to plastic ratio (g solvent/g plastic film)	Temperature (°C)	Time (h)	PE yield (wt %)
1	Filter bag system	<i>n</i> -Dodecane	200	2.5	10.0	95	2	30.02 ± 2.71
2	Soxhlet extraction	Heptane	300	4.0	9.1	96	36	43.36 ± 0.65
3	Jacketed vessel	Dodecane isomers	135	3.6	20.8	95	1	63.82 ± 1.01

One important aspect to consider with a new recycling technology is the potential applications of the recycled materials after their processing. Previous studies have demonstrated some of these with materials that were recovered via dissolution-based approaches. For example, Ramírez-Martínez et al. fabricated membranes using polyolefins that were dissolved in biobased solvents like D-limonene and α -pinene.²⁶ Saleem et al. produced thin films for oil-sorbent pouches with PP that was recovered from disposable face masks.²⁷ In another study, Phan et al. employed solvents to selectively dissolve elastane from different fiber mixtures, which could be incorporated back in other textiles.²⁸ Sriprom et al. produced foam/natural fiber composites with expanded polystyrene (EPS) that was recycled via a dissolution-based process using a mixture of thinner and acetone.²⁹ Polymer solutions can also find applications as paint additives and in glues for different materials like glass, paper, metals, and cloth. This was demonstrated with the dissolution of polyvinyl(butyl) (PVB) in 2-propanol and ethyl acetate, which also allowed for the removal of solid contaminants.³⁰ In general, a product that can be appropriate for the same or similar applications as the original waste would be preferred. This is considered an upcycling process and it is aligned with circular economy principles in which plastics maintain their value.^{31,32} On the other hand, in a downcycling process, the recycled materials do not possess the optimal qualities and must find applications in markets of lower value.^{33–35}

In this study, we aim to validate that polymers recovered by STRAP are suitable for reuse in film manufacturing. In the plastic packaging industry, polymers are converted to different formats depending on their intended use and properties. For example, these can be manufactured into monolayer or multilayer flexible, semirigid, or rigid films, along with bottles, trays, and jars.³⁶ Specifically for plastic films, these are normally manufactured by two techniques: blown and cast film extrusion.³⁷ In the blown film extrusion process, the polymer melt is inflated by air and chilled air is blown on the outside of the film to cool the polymer bubble.³⁸ After reaching the desired temperature, below the melting point of the polymer, the bubble is flattened, cut, and rolled.³⁹ In the cast film process, the polymer melt is extruded through a system of chilled rolls that produce the desired flattened film.^{36,40} In this paper, the potential applications of STRAP polymers in plastic film production were explored. PE was recovered by STRAP from a postindustrial printed multilayer plastic film that was comprised of PE, EVOH, PET, polyurethane (PU) adhesives, and PU-based inks.¹⁷ We have previously recovered PE, EVOH and PET from this material via three processing steps: (1) PE dissolution in *n*-dodecane, (2) EVOH dissolution in a DMSO and water mixture, and (3) ink removal with gamma-valerolactone. Three STRAP methods were considered to increase the production of PE and study the influence of experimental conditions like solvent selection, amount of solvent, and filtration on the film properties.

Since polyolefins are extensively present in the packaging industry due to their clarity, toughness, inertness, and availability, we looked at increasing the recovery of the STRAP PE component to determine its suitability for film production.⁴¹ In our previously published work, *n*-dodecane at 95 °C selectively dissolved the PE component in the presence of the other polymers and PU-based inks in the multilayer material (Figure S1).¹⁷ This solvent was initially selected after our computational modeling tools suggested alkane solvents, primarily heptane and dodecane, as selective PE solvents over the PU inks. This recovered PE was a mixture of LDPE and LLDPE and did not have significant coloration before extrusion and pelletization. The STRAP PE had similar thermal properties and molecular weight values as the corresponding virgin resins.¹⁷ The PE yield in our previous work was 61.78 ± 1.74 wt %, and this required two sequential steps under the same conditions to recover most of the PE and clean the experimental setup.

After the initial STRAP demonstrations, we looked at modifying our experimental setup to increase the amount of PE recovered per experiment. As presented in Figure 1, three different STRAP methods were considered for PE production after its dissolution from the printed multilayer film. The first method employed a filter bag with a pore size of 600 μ m, in which all the plastic contents and dodecane were mixed under high shear with a solvent to plastic ratio of 10:1. In this system, heating occurred from the bottom part of the vessel and the remaining parts were insulated to avoid heat losses. After the dissolution time was completed, the filter bag was lifted to separate the nondissolved contents from the dissolved PE. This resulted in an average PE yield of 30.02 ± 2.71 wt % (Table 1). It was observed that a significant amount of PE precipitated inside the filter bag as soon as it was removed from the solution, in part due to the low solvent to plastic ratio. Additionally, the solvent and dissolved PE would drip slowly out of the filter bag, even with a 600 μ m pore size, most likely due to the high concentration of plastic contents inside the bag. A second step under the same experimental conditions was carried out to recover the PE residue that was left in the system and the yield was improved to 56.5 ± 1.70 wt %, which is closer to the target PE yields. This was an indication that premature PE precipitation should be avoided, either by including less plastic or more solvent in the feed, minimizing temperature gradients, having filter bags with larger pore sizes, or designing a pressurized filtration to achieve the desired separation. These adjustments were considered in the other STRAP methods.

The second method was a modified Soxhlet extraction setup with a heated extraction chamber (96 °C) that included a filter bag with a pore size of 1 μ m, as shown in Figure 1(b). The total time the solvent was kept in the system was 32 h (in a 9.1:1 solvent to plastic ratio) to ensure most of the PE was recovered, with 4 siphonings per hour. However, the maximum PE yield obtained from this approach was only 43.4 ± 0.65 wt %. The extended time and low PE yield shows the disadvantages of this approach for producing the necessary amounts of the desired

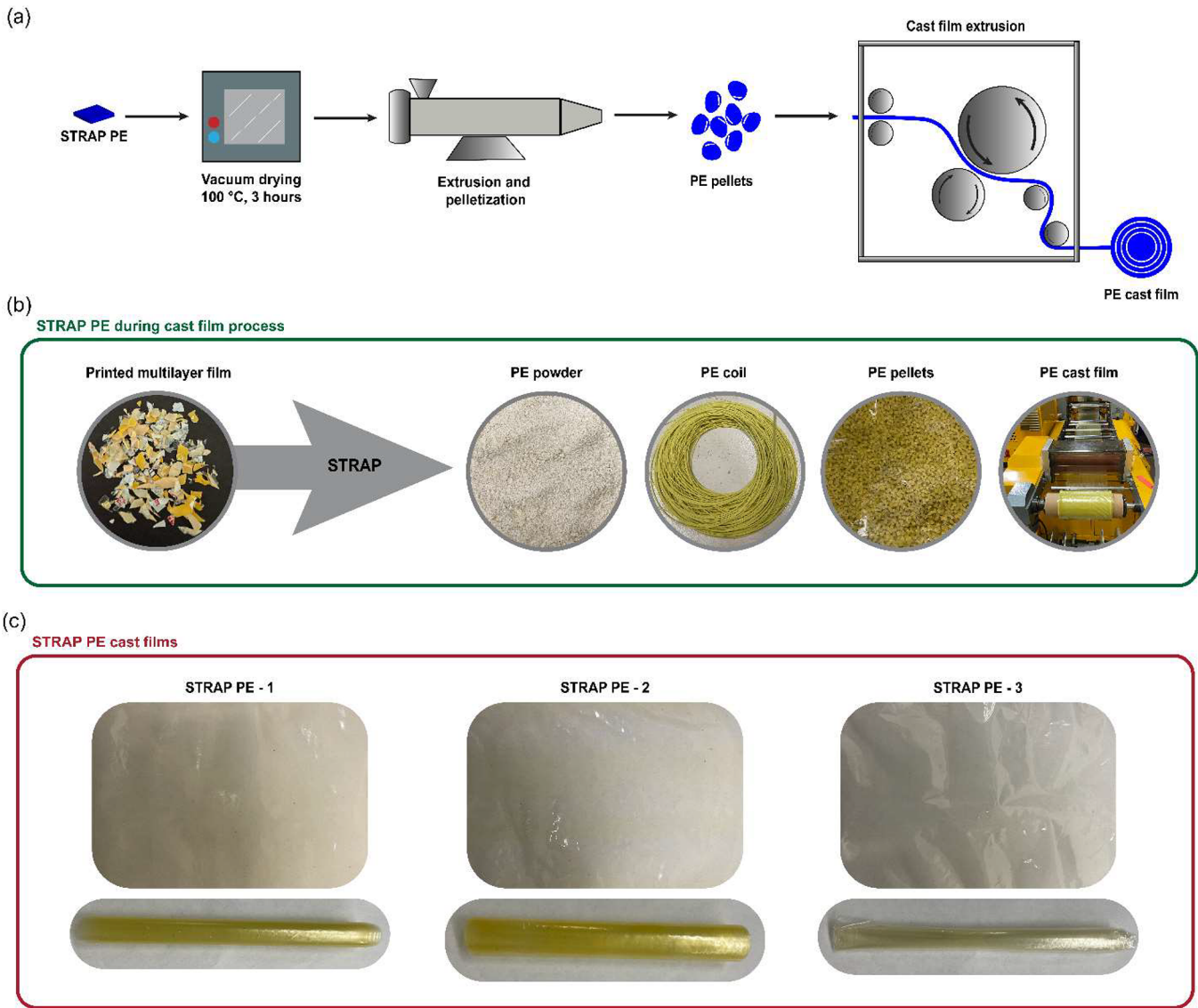


Figure 2. (a) Production of a cast film with polyethylene (PE) recovered from a postindustrial multilayer printed film by solvent-targeted recovery and precipitation (STRAP), (b) images of STRAP PE during the cast film production, and (c) images of the final STRAP PE cast films (each number correspond to the methods listed in Table 1).

Table 2. Tensile Strength Properties of STRAP PE Cast Films^a

Film sample	Film thickness (μm)	Young's modulus (MPa)	Stress at break (MPa)	Force at break (N)	Elongation at break (%)
Machine direction					
C1 ^b	79 $\pm$ 15.08	124 $\pm$ 4.49	32.2 $\pm$ 1.03	64.6 $\pm$ 6.85	719 $\pm$ 38.38%
C2 ^c	82 $\pm$ 14.36	112 $\pm$ 1.12	43.0 $\pm$ 0.64	90.3 $\pm$ 11.85	745 $\pm$ 42.43%
STRAP PE-1	124 $\pm$ 10.40	158 $\pm$ 4.35	14.5 $\pm$ 3.05	57.2 $\pm$ 11.53	494 $\pm$ 109.42%
STRAP PE-2	87 $\pm$ 5.76	108 $\pm$ 6.00	13.1 $\pm$ 0.74	27.6 $\pm$ 5.54	698 $\pm$ 29.53%
STRAP PE-3	84 $\pm$ 12.60	120 $\pm$ 5.73	21.4 $\pm$ 1.09	45.7 $\pm$ 4.62	592 $\pm$ 18.52%
Transverse direction					
C1 ^b	82 $\pm$ 1.98	122 $\pm$ 4.85	24.6 $\pm$ 0.26	55.7 $\pm$ 1.69	854 $\pm$ 7.59%
C2 ^c	74 $\pm$ 3.00	119 $\pm$ 5.37	34.2 $\pm$ 2.02	71.0 $\pm$ 4.53	803 $\pm$ 16.87%
STRAP PE-1	127 $\pm$ 12.95	167 $\pm$ 16.10	12.1 $\pm$ 2.63	47.5 $\pm$ 5.83	568 $\pm$ 86.70%
STRAP PE-2	84 $\pm$ 4.93	103 $\pm$ 7.95	9.1 $\pm$ 0.15	22.0 $\pm$ 1.04	622 $\pm$ 26.95%
STRAP PE-3	81 $\pm$ 4.36	134 $\pm$ 7.10	18.8 $\pm$ 1.04	40.6 $\pm$ 3.40	685 $\pm$ 24.54%

^aFilm sample numbers correspond to the three STRAP methods listed in Table 1. ^bC1 film has LDPE, LLDPE, and tie layers. ^cC2 film has VLDPE, LLDPE, and EVA.

polymers with the STRAP approach. The third setup was a jacketed dissolution tank that had an inner filter with a pore size

of 100 μm (Figure 1(c)). This design allowed for the inclusion of multiple filtration steps within the system, in case a more

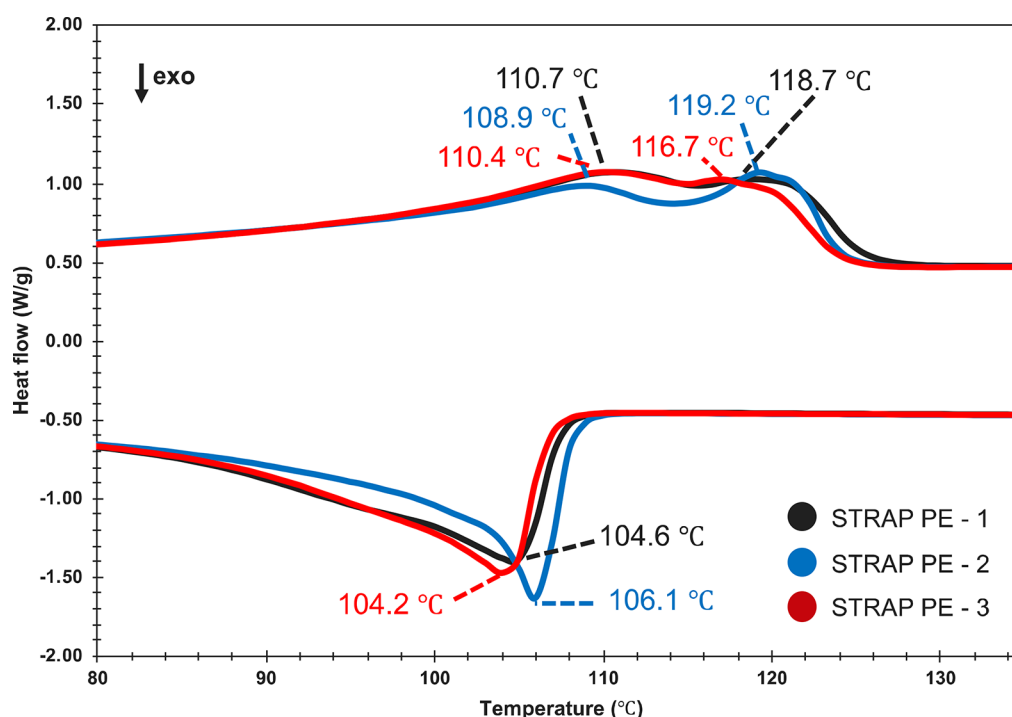


Figure 3. DSC thermogram of the STRAP PE cast films.

rigorous separation would be needed, along with the option for pressurization and optimized heating. Silicone oil was used as a heat transfer fluid in the jacket of the dissolution tank. The solvent and plastics were added in a 20.8:1 ratio and an average PE yield of 63.8 ± 1.01 wt % was obtained, which was comparable to the target value.¹⁷ This result indicated that combining a higher solvent to plastic ratio and improved heating can allow for the optimal recovery of the PE component, even with a filter pore size of 100 μm .

After producing around 1-kg of PE from each method, the polymers were dried, cryo-ground, extruded, and pelletized before the cast film production (Figure 2). Property comparisons between the three cast films and standard LLDPE/LDPE films were made, including color, tensile strength, and number of impurities present in the films. Images of the STRAP PE cast films are presented in Figure 2(c). Additional details about each film can be found in Table S1 in the Supporting Information.

Packaging films must meet different performance criteria according to their intended applications.⁴² This assessment can be made by understanding their mechanical or tensile properties, which can determine how suitable the material is for packaging. Tensile properties will depend on the film manufacturing process, film structure, and testing parameters.⁴³ Regardless, we aimed for an initial analysis of the mechanical performance of the cast films produced from STRAP PE. The tensile properties of the STRAP PE cast films, including Young's modulus, stress at break, force at break, and elongation at break, were compared to two control films that contained LDPE, LLDPE, VLDPE, and EVA (C1 and C2 in Table 2) in both the machine and transverse directions. The machine direction is the direction in which the material comes out of the equipment and the transverse is the perpendicular direction. For flexible packaging, low stiffness is preferred for optimal performance, as it reduces crackling sounds and crazing during processing.⁴⁴ The STRAP PE-1 film (the numbering scheme indicates a film

produced from the corresponding numbered method listed in Table 1) exhibited a stiff behavior, as it possessed the highest Young's modulus and lowest elongation at break out of all the samples, on both the machine and transverse directions. The STRAP PE-1 film also had the largest variability in the elongation at break, possibly caused by the heterogeneity of the film as a result of impurities. It has been observed that the presence of impurities can diminish the mechanical and barrier properties of plastic films since these can disrupt the polymer matrix.⁴⁵ The STRAP PE-2 and 3 films had comparable Young's moduli to the control films; however, lower stresses and forces at break were observed, along with a lower elongation percentage (Table 2).

The STRAP PE films had similar thermal properties, with comparable melting and crystallization temperatures (Figure 3). Two melting temperatures were observed in each sample since the STRAP PE is composed of both LDPE and LLDPE. As seen in Table 3, the films also had comparable heat of melting during the second cycle and percentage crystallinity.

Table 3. Thermal Properties of the STRAP PE Cast Films

Cast film sample	ΔH_c (J/g)	$\Delta H_{m,2}$ (J/g)	Crystallinity (%)
STRAP PE-1	119.0	121.8	42
STRAP PE-2	111.2	114.2	39
STRAP PE-3	116.9	120.0	41

Impurities were detected and counted from images taken in a sample area from the different films. The average number of impurities in the STRAP PE films correlated with the filter pore size in the corresponding STRAP method (Table 4). Finer filter pore sizes reduced the presence of impurities in the final film. The STRAP PE-1 film had the most impurities since a filter with a pore size of 600 μm was used during the STRAP process. The impurities were most likely nondissolved components from the

Table 4. Comparison of Impurities Present in STRAP PE Cast Films

Film sample	STRAP filter pore size (μm)	Average number of impurities per 3600 mm^2
C1 ^a	-	457 $\pm$ 163
C2 ^b	-	591 $\pm$ 266
STRAP PE-1	600	2475 $\pm$ 378
STRAP PE-2	1	798 $\pm$ 237
STRAP PE-3	100	1273 $\pm$ 254

^aC1 film has LDPE, LLDPE, and tie layers. ^bC2 film has VLDPE, LLDPE, and EVA.

multilayer film, EVOH and PET, and this further supports our findings with the mechanical properties of the STRAP-PE 1 film.

The STRAP PE cast films were compared in terms of color, clarity, and haze. As seen in Table 5, an improvement in clarity

Table 5. Clarity and Haze Comparison of the Cast Films^a

Film sample	Clarity	Haze (%)
C1 ^b	87.8	7.9
C2 ^c	94.4	4.7
STRAP PE-1	59.0	30.9
STRAP PE-2	83.4	26.7
STRAP PE-3	86.9	8.1

^aValues presented are an average from 5 separate measurements. ^bC1 film has LDPE, LLDPE, and tie layers. ^cC2 film has VLDPE, LLDPE, and EVA.

and haze was observed between the STRAP PE films, with the STRAP PE-3 film showing comparable clarity and haze to the control films C1 and C2. In the CIELAB color space, the STRAP PE-1 and 2 films had higher b^* and lower a^* values than the control films (Figure 4). This indicates the presence of colors in

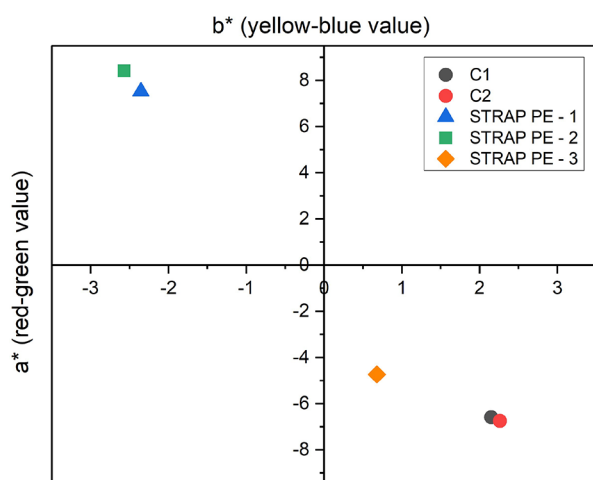


Figure 4. CIELAB color space values of the cast films: a^* is the red-green color value ($a^* > 0$ is redder, $a^* < 0$ is greener), b^* is the yellow-blue color value ($b^* > 0$ yellower, $b^* < 0$ is bluer).

the yellow and green directions. A reduction in the b^* value and increase in a^* value was observed with the STRAP PE-3 film, which had closer color values to the PE control films.

The color improvement between the STRAP PE films was also observed in the yellowness index (YI), with the STRAP PE-3 cast film showing the lowest value among the STRAP films (Figure 5(a)). This reduction in yellow color was attributed to

the amount of solvent used per experiment in the corresponding STRAP system (Figure 5(b)). Applying higher solvent to plastic ratios avoided the accumulation of color bodies in the solvent after each use. A more dilute system also aided in processing challenges in the filtration step for polymer–solvent separation, as it is more difficult to remove the solvent from the polymer in high concentrations.

The solvents from the STRAP methods 1 and 3 were qualitatively analyzed with gas chromatography–mass spectrometry (GCMS) to characterize the accumulation of species after the PE dissolution and recovery. As seen in Figure S2 in the Supporting Information, different plasticizers and slip agents were detected in the post-STRAP n-dodecane from the first method, along with short-chain PE oligomers. The dodecane isomers from the third method did not show significant accumulation of impurities after STRAP (Figure S3). This is because, in the third method, a less concentrated system was used with a higher solvent to plastic ratio. Components related to color were not detected with this method. The buildup of different species in the STRAP solvents is more noticeable with lower plastic to solvent ratios during the polymer dissolution, as was observed with the filter bag method.

The STRAP process has been successfully demonstrated for the recovery of different polymeric components from a flexible multilayer plastic packaging film. Three methods were considered for the recovery of PE from a postindustrial printed multilayer film: a filter bag system, a Soxhlet extraction setup, and a jacketed dissolution vessel. These different systems allowed us to understand how various processing parameters can influence the properties of the cast films produced from the recovered PE. The third method, the jacketed dissolution vessel, provided improvements in color and removal of impurities from the PE. The solvent to plastic ratio proved to be an important parameter, where higher ratios reduced the buildup of impurities like plasticizers, slip agents, and color components. This study demonstrates that STRAP polymers can be recycled back into plastic films, enabling the potential circularity of these packaging materials. Future work will focus on detailed studies on color removal with adsorption beds, characterization of color components in the solvents, and cast film production from other resins recovered by STRAP.

EXPERIMENTAL SECTION

Materials. The printed flexible multilayer film was collected from an Amcor postindustrial waste stream. Average compositions of these materials can be 17 wt % PET, 8 wt % EVOH, 5 wt % inks and adhesives, and 70 wt % PEs. The materials were initially shredded to 3 mm in size through a 1/8" cross-cut shredder (Make: Allegheny 16–75CX) at Michigan Technological University. N-dodecane (ReagentPlus, $\geq 99\%$) was received from Sigma-Aldrich. N-heptane ($\geq 99.3\%$) was received from BBV (Naperville, IL). Dodecane, mixture of isomers, was received from Thermo Scientific Chemicals.

STRAP Experimental Procedure. The STRAP PE recovery step was carried out using three experimental methods with different heating setups and filter sizes. In these setups, 150 to 200 g of printed multilayer film was processed per experiment, with different amounts of a PE-selective solvent. During the polymer dissolution in each method, the dissolved PE was separated from the nondissolved contents. The first method utilized a filter bag with a pore size of 600 μm during the PE dissolution. N-dodecane and the plastic contents were mixed inside the bag with a high shear mixer at 1500 rpm and 95 $^\circ\text{C}$ for

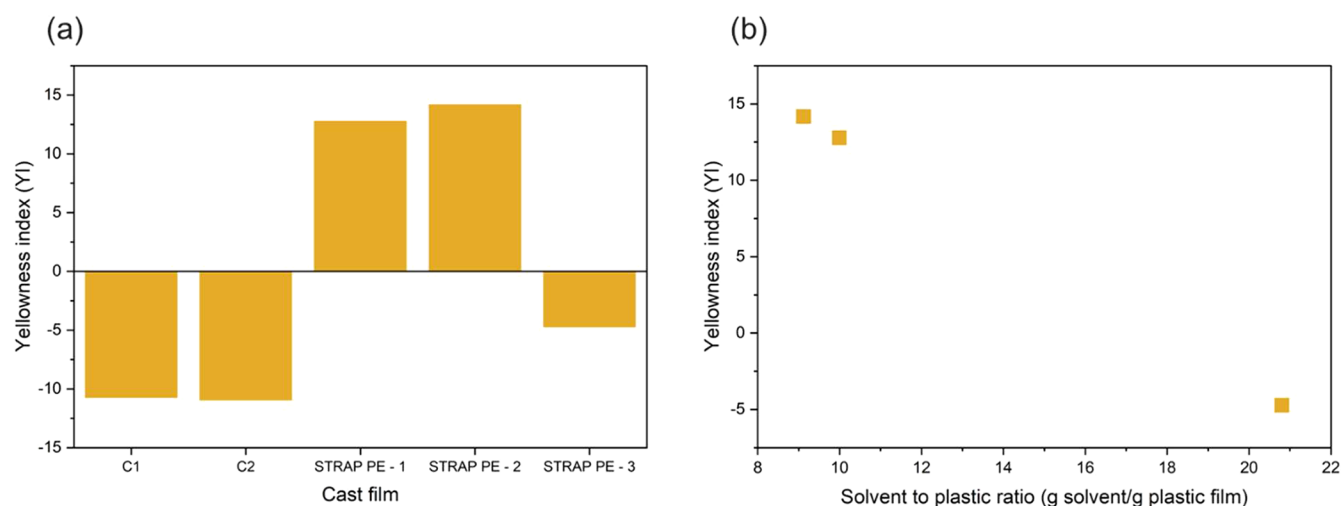


Figure 5. Color comparison of the cast films: (a) yellowness index (YI) of the cast films and (b) YI of the cast films with increasing solvent to plastic ratio in STRAP.

2 h. The second method consisted of a heated Soxhlet extraction system (96 °C) using heptane and a Nomex filter bag with a 1 μm pore size, 32 h of extraction, and 4 siphonings per hour.⁴⁶ In the third method, the plastic contents were put in contact with dodecane isomers in a jacketed dissolution vessel at 95 °C for 1 h. This system used silicone oil as a heat transfer fluid to achieve the desired temperature and allowed for pressurization during filtration. In our test with the dissolution tank, a filter size of 100 μm was used. The recovered PEs from each method were let to precipitate by lowering the temperature of the contents to room temperature. The PE solids were then separated from the solvent by vacuum filtration and later dried for up to 3 h at 100 °C in a vacuum oven to remove residual solvents.

Extrusion of Recovered Polyethylene and Cast Film Production. The recovered PE from the three STRAP methods was cryo-ground with liquid nitrogen in a Thomas Model 4 Wiley Mill to a 3 mm powder before extrusion. The PE powder was extruded to 3 mm pellets using a Leistritz twin-screw extruder with a temperature range of 170–190 °C and a 19-rpm screw speed. After this, the PE pellets were extruded into a cast film at an average barrel temperature of 177 °C and die temperature of 193 °C with a Labtech Engineering cast film extruder.

Characterization of Cast Films. Tensile properties of the films were measured using an Instron testing machine following the ASTM Standard D882–1.⁴⁷ The thermal properties of the films were analyzed with a NETZSCH TGA 209 F1 Libra. The analysis compared the dynamic cooling and the second heating of the films, from –20 to 250 °C, at rate of 10 K/min. The film crystallinity was determined by dividing the second heat of melting by a reference heat of melting for a 100% crystalline linear polyethylene:^{48,49}

$$\text{crystallinity} = \frac{H_{m,2}}{H_m^0} \times 100 (\%)$$

Final color of the cast films was quantified with a spectrophotometer and measurements were converted to CIELAB color space values, L* (lightness), a* (red-green color component) and b* (blue-yellow color component), and YI (yellowness index).⁵⁰ Optical properties were analyzed following the ASTM standard D1003-21 for haze and luminous transmittance of transparent plastics.⁵¹ The defect analysis for

determining the number of impurities in the cast films was performed with an Amcor internal method.

Characterization of STRAP Solvents. The recovered solvents after STRAP were analyzed with gas chromatography–mass spectroscopy (GCMS) to detect species that accumulate in the solvents after STRAP. A Shimadzu GCMS-QP2010 with a DB-5 column (Agilent) at a maximum temperature of 310 °C was used, with helium as the carrier gas.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmaterialslett.4c01048>.

Information about the produced films, STRAP solvent characterization chromatograms, and peak identification (PDF)

■ AUTHOR INFORMATION

Corresponding Author

George W. Huber – Department of Chemical and Biological Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States; orcid.org/0000-0002-7838-6893; Email: gwhuber@wisc.edu

Authors

Kevin L. Sánchez-Rivera – Department of Chemical and Biological Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States

Charles Granger – Department of Chemical and Biological Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States; orcid.org/0000-0002-0160-7520

Harrison Appiah – Department of Forest, Rangeland and Fire Sciences, University of Idaho, Moscow, Idaho 83844, United States; orcid.org/0000-0002-1101-7225

Kevin Nelson – Amcor, Neenah Innovation Center, Neenah, Wisconsin 54956, United States

Steve Grey – Amcor, Neenah Innovation Center, Neenah, Wisconsin 54956, United States

David J. Sun – Department of Chemical and Biological Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States

John E. Estela-García – Department of Chemical and Biological Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States

Edward Chen – Department of Mechanical Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States; Wisconsin Institute for Discovery, University of Wisconsin-Madison, Madison, Wisconsin 53715, United States

Zhuo Xu – Department of Chemical and Biological Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States

Tim A. Osswald – Department of Mechanical Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States

Lih-Sheng Turng – Department of Mechanical Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States; Wisconsin Institute for Discovery, University of Wisconsin-Madison, Madison, Wisconsin 53715, United States; orcid.org/0000-0001-8022-9224

Armando G. McDonald – Department of Forest, Rangeland and Fire Sciences, University of Idaho, Moscow, Idaho 83844, United States

Reid C. Van Lehn – Department of Chemical and Biological Engineering, University of Wisconsin-Madison, Madison, Wisconsin 53706, United States; orcid.org/0000-0003-4885-6599

Ezra Bar-Ziv – Department of Mechanical Engineering, Michigan Technological University, Houghton, Michigan 49931, United States; orcid.org/0000-0003-1444-123X

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acsmaterialslett.4c01048>

Author Contributions

CRediT: **Kevin L. Sánchez-Rivera** conceptualization, data curation, formal analysis, investigation, methodology, writing-original draft, writing-review & editing; **Charles Granger** conceptualization, data curation, formal analysis, investigation, methodology, writing-review & editing; **Harrison Appiah** investigation, methodology, writing-review & editing; **Kevin Nelson** data curation, investigation, writing-review & editing; **Steven Grey** investigation, writing-review & editing; **David J. Sun** data curation, investigation, writing-review & editing; **Edward Chen** investigation, writing-review & editing; **Z. Xu** conceptualization, writing-review & editing; **Tim Osswald** conceptualization, writing-review & editing; **Armando G. McDonald** conceptualization, data curation, writing-review & editing; **Reid C. Van Lehn** conceptualization, funding acquisition, writing-review & editing; **Ezra Bar-Ziv** conceptualization, writing-review & editing; **George W. Huber** conceptualization, formal analysis, funding acquisition, project administration, supervision, writing-review & editing.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This material is based upon work supported by the U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, Bioenergy Technologies Office under Award Number DE-EE0009285. This research was partially supported by NETZSCH using instrumentation provided at the University of Wisconsin-Madison Polymer Engineering Center.

REFERENCES

- (1) Geyer, R.; Jambeck, J. R.; Law, K. L. Production, use, and fate of all plastics ever made. *Science Advances* **2017**, 3, No. e1700782.
- (2) Grabel, T.; Dixon, C.; Kean-Hammerson, J.; Gammage, T. *On the Road to Success: Designing an Effective Plastics Treaty*; Environmental Investigation Agency, UN Environment Programme, 2022.
- (3) Heller, M. C.; Mazor, M. H.; Keoleian, G. A. Plastics in the US: toward a material flow characterization of production, markets and end of life. *Environmental Research Letters* **2020**, 15, 094034.
- (4) Niaounakis, M. *Recycling of Flexible Plastic Packaging*; Elsevier, 2020. DOI: [10.1016/C2018-0-00076-6](https://doi.org/10.1016/C2018-0-00076-6).
- (5) Bertolucci, G.; Leroy, Y.; Olsson, A. Exploring the environmental impacts of olive packaging solutions for the European food market. *Journal of Cleaner Production* **2014**, 64, 234–243.
- (6) O'Farrell, K.; Lewis, H. *Design Smart Material Guide—Flexible Plastic Packaging*; Australian Packaging Covenant, 2013.
- (7) Horodytska, O.; Valdés, F. J.; Fullana, A. Plastic flexible films waste management - A state of art review. *Waste Management* **2018**, 77, 413–425.
- (8) Coles, R.; Kirwan, M., Eds. *Food and Beverage Packaging Technology*; Blackwell Publishing, Ltd., 2011. DOI: [10.1002/9781444392180](https://doi.org/10.1002/9781444392180).
- (9) Anukiruthika, T.; Sethupathy, P.; Wilson, A.; Kashampur, K.; Moses, J. A.; Anandharamakrishnan, C. Multilayer packaging: Advances in preparation techniques and emerging food applications. *Comprehensive Reviews in Food Science and Food Safety* **2020**, 19, 1156–1186.
- (10) Tartakowski, Z. Recycling of packaging multilayer films: New materials for technical products. *Resources, Conservation and Recycling* **2010**, 55, 167–170.
- (11) Kaiser, K.; Schmid, M.; Schlummer, M. Recycling of Polymer-Based Multilayer Packaging: A Review. *Recycling* **2018**, 3, 1.
- (12) Sánchez-Rivera, K. L.; Zhou, P.; Kim, M. S.; González Chávez, L. D.; Grey, S.; Nelson, K.; Wang, S.-C.; Hermans, I.; Zavala, V. M.; Van Lehn, R. C.; Huber, G. W. Reducing Antisolvent Use in the STRAP Process by Enabling a Temperature-Controlled Polymer Dissolution and Precipitation for the Recycling of Multilayer Plastic Films. *ChemSusChem* **2021**, 14, 4317–4329.
- (13) Walker, T. W.; Frelka, N.; Shen, Z.; Chew, A. K.; Banick, J.; Grey, S.; Kim, M. S.; Dumesic, J. A.; Van Lehn, R. C.; Huber, G. W. Recycling of multilayer plastic packaging materials by solvent-targeted recovery and precipitation. *Science Advances* **2020**, 6, No. eaba7599.
- (14) Zhou, P.; Sánchez-Rivera, K. L.; Huber, G. W.; Van Lehn, R. C. Computational Approach for Rapidly Predicting Temperature-Dependent Polymer Solubilities Using Molecular-Scale Models. *ChemSusChem* **2021**, 14, 4307–4316.
- (15) Zhou, P.; Yu, J.; Sánchez-Rivera, K. L.; Huber, G. W.; Van Lehn, R. C. Large-scale computational polymer solubility predictions and applications to dissolution-based plastic recycling. *Green Chem.* **2023**, 25, 4402–4414.
- (16) Munguía-López, A. D. C.; Göreke, D.; Sánchez-Rivera, K. L.; Aguirre-Villegas, H. A.; Avraamidou, S.; Huber, G. W.; Zavala, V. M. Quantifying the environmental benefits of a solvent-based separation process for multilayer plastic films. *Green Chem.* **2023**, 25, 1611–1625.
- (17) Sánchez-Rivera, K. L.; Munguía-López, A. D. C.; Zhou, P.; Cecon, V. S.; Yu, J.; Nelson, K.; Miller, D.; Grey, S.; Xu, Z.; Bar-Ziv, E.; Vorst, K. L.; Curtzwiler, G. W.; Van Lehn, R. C.; Zavala, V. M.; Huber, G. W. Recycling of a post-industrial printed multilayer plastic film containing polyurethane inks by solvent-targeted recovery and precipitation. *Resources, Conservation and Recycling* **2023**, 197, 107086.
- (18) Sánchez-Rivera, K. L. *Recycling of Multilayer Plastic Packaging Materials by Solvent Targeted-Recovery and Precipitation*; University of Wisconsin-Madison, 2024.
- (19) Ugduler, S.; Van Laere, T.; De Somer, T.; Gusev, S.; Van Geem, K. M.; Kulawig, A.; Leineweber, R.; Defoin, M.; Van den Bergen, H.; Bontinck, D.; De Meester, S. Understanding the complexity of deinking plastic waste: An assessment of the efficiency of different treatments to remove ink resins from printed plastic film. *J. Hazard. Mater.* **2023**, 452, 131239.

- (20) Das, P.; Zeng, Q.; Leybros, A.; Gabriel, J.-C. P.; Yong Tay, C.; Lee, J.-M. Enhanced extraction of brominated flame retardants from e-waste plastics. *Chem. Eng. J.* **2023**, *469*, 144126.
- (21) Ferreira, A.M.; Sucena, I.; Otero, V.; Angelin, E.M.; Melo, M.J.; Coutinho, J. A. P. Pretreatment of Plastic Waste: Removal of Colorants from HDPE Using Biosolvents. *Molecules* **2022**, *27*, 98.
- (22) Van Melkebeke, M.; De Somer, T.; Van Laere, T.; Nguyen Luu Minh, T.; Mohamadzadeh Shirazi, H.; Poelman, H.; Van Geem, K.; De Meester, S. Adsorption modeling for contaminant removal in plastic dissolution recycling: Investigating an amino ketone-based red dye. *Sep. Purif. Technol.* **2024**, *331*, 125559.
- (23) Roosen, M.; Harinck, L.; Ügdüler, S.; De Somer, T.; Hucks, A.-G.; Belé, T. G. A.; Buettner, A.; Ragaert, K.; Van Geem, K. M.; Dumoulin, A.; De Meester, S. Deodorization of post-consumer plastic waste fractions: A comparison of different washing media. *Science of The Total Environment* **2022**, *812*, 152467.
- (24) Mumbach, G. D.; de Sousa Cunha, R.; Machado, R. A. F.; Bolzan, A. Dissolution of adhesive resins present in plastic waste to recover polyolefin by sink-float separation processes. *Journal of Environmental Management* **2019**, *243*, 453–462.
- (25) Accelerating circular supply chains for plastics: A landscape of transformational technologies that stop plastic waste, keep materials in pay and grow markets; Closed Loop Partners, 2019.
- (26) Ramírez-Martínez, M.; Aristizábal, S. L.; Szekely, G.; Nunes, S. P. Bio-based solvents for polyolefin dissolution and membrane fabrication: from plastic waste to value-added materials. *Green Chem.* **2023**, *25*, 966–977.
- (27) Saleem, J.; Moghal, Z.K.; Shakoor, R.A.; McKay, G. Sustainable Solution for Plastic Pollution: Upcycling Waste Polypropylene Masks for Effective Oil-Spill Management. *International Journal of Molecular Sciences* **2023**, *24*, 12368.
- (28) Phan, K.; Ügdüler, S.; Harinck, L.; Denolf, R.; Roosen, M.; O'Rourke, G.; De Vos, D.; Van Speybroeck, V.; De Clerck, K.; De Meester, S. Analysing the potential of the selective dissolution of elastane from mixed fiber textile waste. *Resources, Conservation and Recycling* **2023**, *191*, 106903.
- (29) Sriprom, W.; Sirivallop, A.; Choodum, A.; Limsakul, W.; Wongniramaikul, W. Plastic/Natural Fiber Composite Based on Recycled Expanded Polystyrene Foam Waste. *Polymers* **2022**, *14*, 2241.
- (30) Królikowski, M.; Żach, P.; Kalestyński, M. Selection of Conditions in PVB Polymer Dissolution Process for Laminated Glass Recycling Applications. *Polymers* **2022**, *14*, 5119.
- (31) Horodytska, O.; Kiritsis, D.; Fullana, A. Upcycling of printed plastic films: LCA analysis and effects on the circular economy. *Journal of Cleaner Production* **2020**, *268*, 122138.
- (32) Campbell-Johnston, K.; Vermeulen, W. J. V.; Reike, D.; Brulot, S. The Circular Economy and Cascading: Towards a Framework. *Resources, Conservation & Recycling: X* **2020**, *7*, 100038.
- (33) Eriksen, M. K.; Damgaard, A.; Boldrin, A.; Astrup, T. F. Quality Assessment and Circularity Potential of Recovery Systems for Household Plastic Waste. *Journal of Industrial Ecology* **2019**, *23*, 156–168.
- (34) Sung, K. A review on upcycling: current body of literature, knowledge gaps and a way forward. *The ICECESS 2015: 17th International Conference on Environmental, Cultural, Economic and Social Sustainability*, Venice, Italy, 2015.
- (35) van der Harst, E.; Potting, J.; Kroeze, C. Comparison of different methods to include recycling in LCAs of aluminium cans and disposable polystyrene cups. *Waste Management* **2016**, *48*, 565–583.
- (36) Riley, A. Plastics manufacturing processes for packaging materials. *Packaging Technology: Fundamentals, Materials and Processes* **2012**, 310–360.
- (37) Rosato, D. V.; Rosato, D. V. Extrusion. In *Plastics Processing Data Handbook*; Rosato, D. V., Rosato, D. V., Eds.; Springer Netherlands: Dordrecht, 1990; pp 91–169.
- (38) Selke, S. E.; Hernandez, R. J. Packaging: Polymers in Flexible Packaging. In *Encyclopedia of Materials: Science and Technology*; Buschow, K. H. J., Cahn, R. W., Flemings, M. C., Ilshner, B., Kramer, E. J., Mahajan, S., Veyssière, P., Eds.; Elsevier: Oxford, 2001; pp 6652–6656.
- (39) Crawford, R. J.; Martin, P. J. Processing of plastics. In *Plastics Engineering*, 4th ed.; Crawford, R. J., Martin, P. J., Eds.; Butterworth-Heinemann, 2020; Chapter 4, pp 279–409.
- (40) Billham, M.; Clarke, A. H.; Garrett, G.; McNally, G. M.; Murphy, W. R. The Effect of Extrusion Processing Conditions on the Properties of Blown and Cast Polyolefin Packaging Films. *Developments in Chemical Engineering and Mineral Processing* **2003**, *11*, 137–146.
- (41) Majumder, K. K.; Hobbs, G.; Bhattacharya, S. N. Molecular, rheological, and crystalline properties of low-density polyethylene in blown film extrusion. *Polym. Eng. Sci.* **2007**, *47*, 1983–1991.
- (42) Krishnaswamy, R. K.; Lamborn, M. J. Tensile properties of linear low density polyethylene (LLDPE) blown films. *Polym. Eng. Sci.* **2000**, *40*, 2385–2396.
- (43) Crawford, R. J.; Martin, P. J. Mechanical behaviour of plastics. In *Plastics Engineering*, 4th ed.; Crawford, R. J., Martin, P. J., Eds.; Butterworth-Heinemann, 2020; Chapter 2, pp 59–194.
- (44) Curtzweiler, G.; Vorst, K.; Palmer, S.; Brown, J. W. Characterization of Current Environmentally-Friendly Films. *Journal of Plastic Film & Sheet* **2008**, *24*, 213–226.
- (45) Lopes, A. C.; Barcia, M. K.; Veiga, T. B.; Yamashita, F.; Grossmann, M. V. E.; Olivato, J. B. Eco-friendly materials produced by blown-film extrusion as potential active food packaging. *Polym. Adv. Technol.* **2021**, *32*, 779–788.
- (46) Abbasi, M.; Coats, E. R.; McDonald, A. G. Green solvent extraction and properties characterization of Poly(3-hydroxybutyrate-co-3-hydroxyvalerate) biosynthesized by mixed microbial consortia fed fermented dairy manure. *Bioresource Technology Reports* **2022**, *18*, 101065.
- (47) Standard Test Method for Tensile Properties of Thin Plastic Sheet, ASTM Standard D882-1; ASTM International, 2018.
- (48) Cecon, V. S.; Curtzweiler, G. W.; Vorst, K. L. A Study on Recycled Polymers Recovered from Multilayer Plastic Packaging Films by Solvent-Targeted Recovery and Precipitation (STRAP). *Macromol. Mater. Eng.* **2022**, *307*, 2200346.
- (49) Thomas, L. Characterization of melting phenomena in linear low density polyethylene by modulated DSC; TA Instruments, Inc.
- (50) Standard Test Method for Color Determination of Plastic Pellets, ASTM D6290-19; ASTM International, 2019.
- (51) Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics, ASTM D1003-21; ASTM International, 2021.